Sravani Ganta

sravaniganta1999@gmail.com | +1 (226)-972-3043 | linkedin | Github

Summary

Skilled Azure Cloud & Data Automation Engineer with 5+ years of experience in designing data pipelines, automating workflows, and operationalizing AI tools in cloud environments. I specialize in Azure Data Factory, Logic Apps, Function Apps, Azure SQL, and Power Platform integrations. Proven success in implementing AI-driven process improvements, REST API integrations, and observability solutions across enterprise applications. Adept at aligning cloud strategies with business goals to improve governance, scalability, and efficiency.

Experience

LTIMindtree, Canada **April 2024 – Present**

Role: Cloud Engineer

- Designed modular ETL pipelines using Azure Data Factory and PySpark in Databricks to ingest structured/unstructured data into Azure Data Lake Storage Gen2.
- Built scalable analytics workflows on Databricks and Synapse using dimensional modeling for downstream AI and BI needs.
- Developed NLP pipelines with Azure OpenAI and Cognitive Search, to enhance semantic document retrieval.
- Automated model training and deployment using Azure ML SDK & CLI, streamlined data prep and hyperparameter tuning.
- Deployed chatbot workflows in Azure Prompt Flow, improving domain-specific accuracy by 20%.
- Engineered solutions with Azure AI Foundry to support digital transformation aligned with business KPIs.
- Containerized ML applications using Docker and deployed on Azure Web Apps, integrating APIs and caching for performance.
- Collaborated on Agile teams to deliver end-to-end AI/ML workflows with data governance and performance optimization.
- Conducted data extraction and transformation using advanced SQL and Python, integrating disparate sources into analytics platforms.

University of Waterloo **September 2023 – December 2023**

Role: Teaching Assistant, Supervisor: Prof. Abdalla Mohamed Hussein, Course: ECE 358 - Computer Networks

- Provided guidance and assistance to undergraduate students in problem-solving and assignments
- Conducted online office hours and offered support to students enrolled in the course

University of Waterloo **April 2023 – August 2023**

Role: Machine learning Researcher, Supervisor: Prof. En Hui Yang

- Designed and implemented weight initialization strategies to transfer knowledge from larger models to smaller models, utilizing pre-trained ResNet and ShuffleNet architectures coupled with Kaiming normalization on CIFAR-100 and ImageNet datasets.
- Achieved 10% faster convergence and a 0.5% increase in accuracy for child models through optimized strategies.

Accenture Private Ltd, India **October 2020 – August 2022**

Role: Database Administrator

- Installed, configured, and upgraded Microsoft SQL Server in complex on-premises and clustered environments, ensuring high availability, scalability, and optimal system reliability.
- Migrated SQL Server databases across environments with minimal downtime, ensuring data integrity and reliability.
- Designed and implemented High Availability and Disaster Recovery solutions, including Database Mirroring, Log Shipping, Transactional Replication, and SQL Server failover clusters, minimizing data loss and enhancing disaster recovery processes.
- Configured and maintained SQL Server maintenance plans to address fragmentation and disk space issues, ensuring successful execution of backups, index rebuilds, and statistics updates through proactive monitoring.
- Managed SQL Server authentication, roles, and permissions, implementing organizational security policies, and configured Linked Servers, Endpoints, Operators, and Database Email for seamless integration and proactive alerting across environments.
- Provided 24/7 on-call support for critical database environments, ensuring stability and availability by conducting consistency checks and implementing granular backup and recovery strategies.

Skills

- **Languages**: Python, SQL, C, ASP.Net, KQL, HTML
- **Tools**: Tableau, Power BI, Docker, Git, Visual Studio, Jupyter
- **Azure Services:** Data Factory, Logic Apps, Function Apps, Azure SQL, Monitor, Application Insights
- **AI & Analytics:** Azure OpenAI, Cognitive Services, Copilot integrations, NLP, ML Pipelines, Prompt Flow
- **Data Processing:** Data Wrangling, Cleaning, Manipulation, Exploration, and Visualization
- **Databases:** Microsoft SQL server, DBA, Database Management systems, RDBMS
- **Data Engineering:** ETL, Dimensional Modeling, Azure Data Lake, Databricks, Synapse, SQL Server
- **CI/CD & Monitoring:** Azure DevOps Pipelines, GitHub, Application Insights, Log Analytics
- **Governance & Security:** IAM, RBAC, Compliance, Policy Enforcement
- **Libraries &frameworks:** TensorFlow, PyTorch, Keras, NumPy, Pandas, PySpark, NLTK, NLP

Education

University of Waterloo **September 2022 – December 2023**
Master of Engineering in Electrical and Computer Engineering

Jawaharlal Nehru Technological University **June 2016 – August 2020**
Bachelor of Technology in Electrical and Instrumentation Engineering

Projects

SQL Migration, Pipeline Dynamics, and Power BI Insights | *Microsoft SQL Server, Microsoft Azure, Data Visualization, ETL, Power BI*

- Orchestrated the migration of an on-premises SQL database to Azure, ensuring seamless data integration from various sources including CSV files. This foundational work laid the groundwork for advanced data manipulation and analysis.
- Engineered a robust ETL pipeline leveraging Azure Data Factory, facilitating the efficient transfer of data from on-premises environments to Azure Storage. This included the creation of Azure resources and the configuration of a self-hosted integration runtime for cloud connectivity.
- Implemented Azure Databricks for enhanced data processing, employing Spark SQL for table aggregation. This step was crucial for handling large volumes of data and performing complex computations efficiently.
- Finalized the data processing framework with the integration of Power BI, enabling sophisticated data visualization and insights. This allowed for the dynamic representation of data analytics, aiding in decision-making processes.

Admin Login page for Chatbot Management| *React, Next.js, JavaScript, Tailwind CSS*

- Developed an admin page using Next.js and React to efficiently manage chatbot user information, including user activity, last seen status, payment dates, and renewal countdowns.
- Designed and implemented an intuitive user interface for admin operations.
- Successfully fetched and displayed user data, formatted for readability.
- Calculated and displayed renewal countdown based on payment dates.

Medical Condition Classification and Drug Review Analysis | *NLP, python, sentiment analysis*

- Developed an NLP and ML-based drug recommendation system using trigram TF-IDF vectorization and the Passive Aggressive Classifier, achieving a 98.2% accuracy rate in evaluations with a drug dataset. This involved preprocessing text data by removing HTML, special characters, and stopwords, and applying stemming and lemmatization.
- Implemented a comprehensive text analysis approach, utilizing both bag of words and TFIDF vectorizer techniques, and integrated Naive Bayes and Passive Aggressive Classifier models for sentiment analysis and drug recommendation.

Google Cloud Data Pipelining with Airflow | *Google API, Airflow, AWS, EC2, python*

- Developed and automated a data pipeline to extract, transform, and clean YouTube video reviews using Google API, Python, and Apache Airflow, storing data securely on Amazon S3.
- Designed and implemented ETL processes to extract data from multiple sources, transform it using custom Python scripts, and load it into a centralized data warehouse.
- Deployed the pipeline on AWS EC2 instances, ensuring high availability and scalability to handle large volumes of data.
- Implemented monitoring and alerting mechanisms using AWS CloudWatch and Airflow's built-in features to ensure the reliability and performance of the data pipeline.
- Automated the data pipeline with Apache Airflow and stored the processed data securely on Amazon S3.

Clustering and Classification with Deep Learning | *Machine Learning, Data Visualization, Deep Neural Networks, Clustering*

- Designed, implemented, and compared multiple deep learning architectures for data classification, including a default architecture and a customized architecture utilizing various Deep Neural Network (DNN) variants. This process involved comprehensive training and evaluation to determine the optimal model for the dataset.
- Conducted advanced clustering techniques to analyze and segment datasets, creating intuitive visualizations to communicate patterns and insights effectively. Leveraged tools like Matplotlib, Seaborn, and Plotly to enhance interpretability and aid in data-driven decision-making.
- Implemented diverse machine learning algorithms, including Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), Naïve Bayes, Decision Trees, Random Forests, and Gradient Tree Boosting, to classify datasets with high accuracy. Developed models to address specific data challenges and improve outcomes.

Certifications

- Azure OpenAI Specialist
- Azure Cognitive Services Specialist
- Machine learning LLM
- AZAI Gen AI Specialist
- AZAI AzureML Advanced
- AZ-303
- Microsoft SQL